Paleomagnetism of fluorite veins in the Devonian St. Lawrence granite, Newfoundland, Canada

K. Kawasaki and D.T.A. Symons

Abstract: Paleomagnetic results are reported from the Devonian St. Lawrence granite and the fluorite veins within the granite or its related porphyry dikes. Paleomagnetic analysis of 359 specimens from 29 sites shows that there is no statistically significant difference at 95% confidence between the site mean characteristic remanent magnetization (ChRM) directions carried by pyrrhotite and (or) magnetite of the granite and fluorite veins. No post-granite metamorphic event is known in the area. Thus, these concordant directions indicate that the granite and fluorite veins are coeval and retain a primary remanence. However, the mean ChRM direction of 25 vein and granite sites yields an Early Pennsylvanian paleopole of ~316 Ma on the North American apparent polar wander path at 35.5°N, 129.2°E (A_{95} = 3.4°) that is younger than the 374 ± 2 Ma Devonian U–Pb zircon age for the granite. Two intersecting vertical vein sets indicate that the St. Lawrence granite has not been tilted since emplacement of the fluorite veins. Therefore, the difference between the measured and expected paleopoles is interpreted to result from a postemplacement counterclockwise rotation of ~17° of the St. Lawrence granite region about a vertical axis. Analysis of Carboniferous paleopoles in the northern Appalachian orogen indicates the rotation at the St. Lawrence area is Late Devonian during the Acadian orogeny.

Résumé : Des résultats paléomagnétiques sont présentés pour le granite St. Lawrence et les filons de fluorine dans le granite ou les dykes de porphyres qui y sont associés. Une analyse paléomagnétique de 359 échantillons provenant de 29 sites montre qu'il n'y a pas de différence statistiquement significative, à un taux de confiance de 95 %, entre les directions moyennes d'aimantation rémanente caractéristique « ChRM » mesurées sur la pyrrhotite et(ou) la magnétite du granite et des filons de fluorine. Aucun événement métamorphique post-granite n'est connu dans le secteur. Ainsi, ces directions concordantes indiquent que le granite et les filons de fluorine sont contemporains et conservent une rémanence primaire. Cependant, la direction ChRM moyenne de 25 sites de granites et de filons donne un paléopôle, d'environ 316 Ma, sur le chemin de dérive des pôles à 35,5°N, 129,2°E (A_{95} = 3,4°), c.-à-d. à la base du Pennsylvanien, ce qui est plus jeune que l'âge U–Pb sur zircon de 374 ± 2 Ma pour le granite, c.-à-d. un âge Dévonien. Deux ensembles de filons s'entrecoupant indiquent que le granite St. Lawrence n'a pas été basculé depuis l'intrusion des filons de fluorine. Donc, la différence entre les paléopôles mesurés et prévus est interprétée en tant que le résultat d'une rotation de la région du granite St. Lawrence; cette rotation d'environ 17° se serait produite après l'intrusion et dans le sens anti-horaire, autour d'un axe vertical. Une analyse des paléopôles au Carbonifère dans l'orogène des Appalaches du Nord indique que la rotation de la région du St. Lawrence s'est produite durant l'orogène acadien au Dévonien tardif.

[Traduit par la Rédaction]

Introduction

Fluorite deposits are formed by various processes under different physicochemical conditions in a range of geologic settings. These deposits are mostly of two lithologic types: (1) those in sedimentary rocks, dominantly carbonates; and (2) those found as veins in, or associated with, dominantly alkaline–peralkaline igneous rocks that are akin to magmatic-hydrothermal vein-type deposits (Strong et al. 1984). More than 40 distinct fluorite veins are known in the St. Lawrence area, Newfoundland (Figs. 1, 2) and most are hosted by the 374 ± 2 Ma St. Lawrence granite and its outliers (Howse et al. 1983). The fluorite veins constitute a major high-grade resource by world standards and are the premier known resource in Canada. During 44 years of operation until closure in 1978, the St. Lawrence fluorspar mines produced >4.2 Mt of ore, and the Director, Blue Beach, and Tarefare veins alone still contain ~8.09 Mt of fluorspar mineralization (Howse et al. 1983).

As well as the spatial relationship, there is a presumed genetic relationship based on geochemical arguments between the St. Lawrence granite and fluorite veins (Teng and Strong 1976). Irving and Strong (1985) reported paleomagnetic data for the St. Lawrence granite but not for the fluorite veins. They observed two remanence components, N and H, in the granite, interpreting the N component to be the primary magnetization acquired during cooling of the pluton and the H component to be a Kiaman overprint from low tempera-

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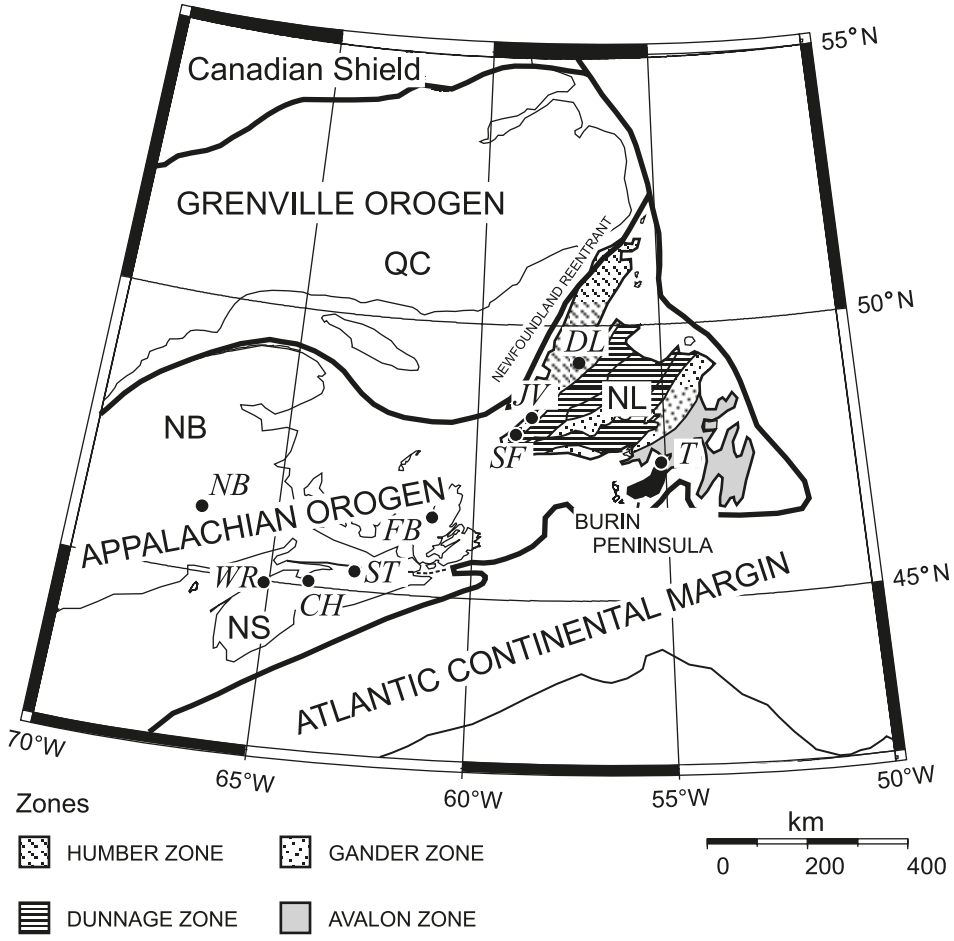
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Fig. 1. Major tectonic elements of Atlantic Canada (modified from Williams et al. 1999). NB, New Brunswick; NL, Newfoundland and Labrador; NS, Nova Scotia; QC, Quebec. Italic letter symbols show sampling locations for Early Carboniferous rocks listed in Table 2.



ture alteration that deposited hematite diffusely throughout the granite. Our study aimed to date the fluorite veins and to establish their genetic relationship to the granite.

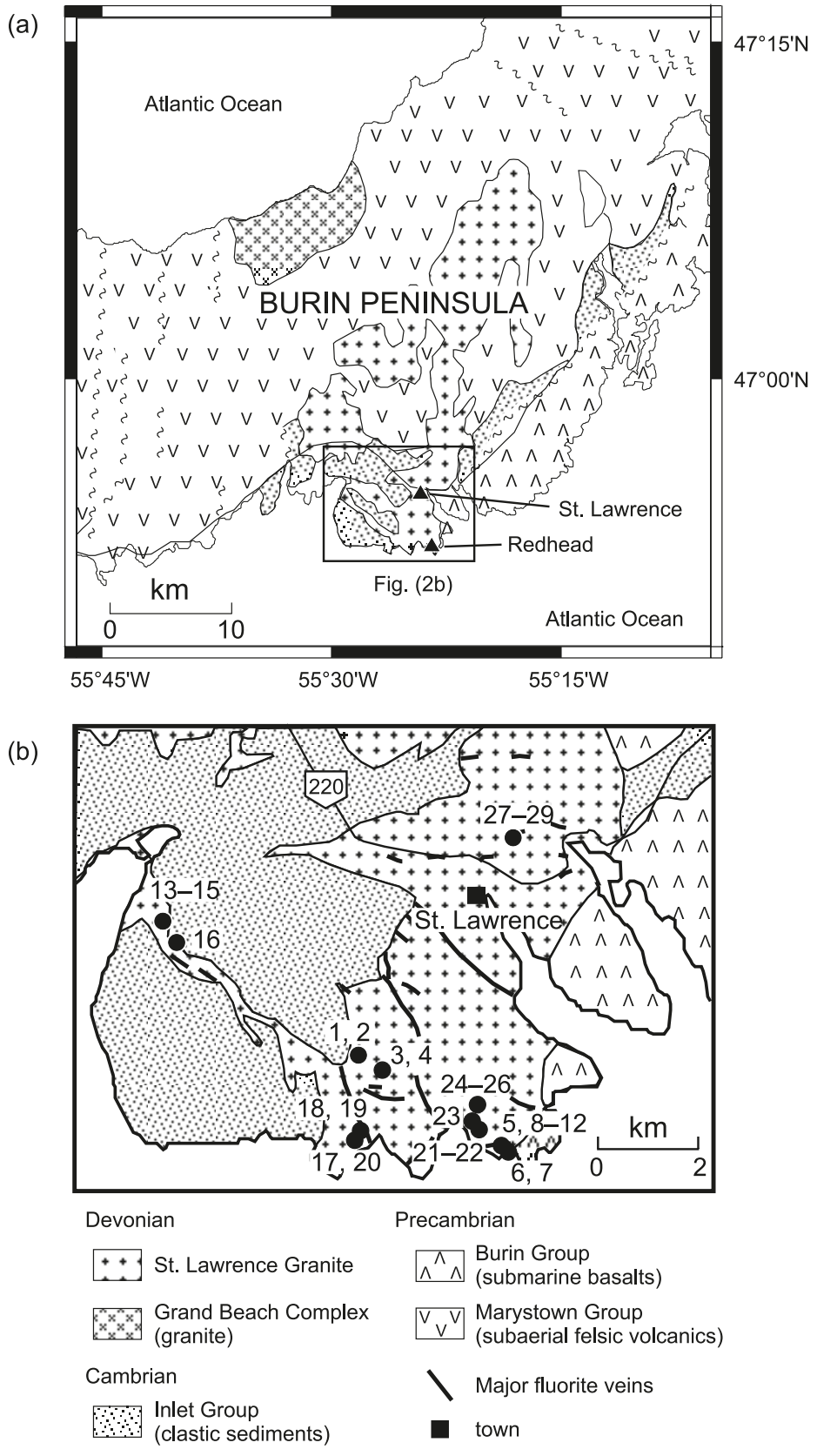
Geology

The St. Lawrence area is in the Avalon terrane (Fig. 1), the widest and most continuous tectonostratigraphic zone of the Appalachians (Williams 1979). The Avalon zone consists mainly of Late Precambrian volcanic and sedimentary rocks that are relatively unmetamorphosed and undeformed compared with coeval rocks of nearby terranes, suggesting that the St. Lawrence granite was intruded into a relatively stable tectonic environment (Collins and Strong 1988). The granite batholith outcrops over an area of ~30 km by 10 km with a number of outliers (Teng and Strong 1976; Fig. 2*a*). This pluton is mainly syenitic to alaskitic in composition, fine to medium grained, porphyritic to equigranular, and bright red in colour (Howse et al. 1983). The granite consists of quartz, orthoclase, and albite with minor amounts of riebeckite, aegirine, biotite, fluorite, magnetite, and hematite. Alkali feldspar makes up 30%–60% of the rock and is generally turbid because of finely disseminated hematite (Irving and Strong 1985). The pluton is clearly post-tectonic, intruding most stratigraphic units in the southern Burin Peninsula and cutting all deformation features such as cleavage, folds, and thrust faults (Strong et al. 1978). Miarolitic cavities occur near the pluton's contacts where tuffisites (gas breccias) are common, indicating shallow emplacement (Irving and Strong 1985). Kerr et al. (1993) reported a U–Pb zircon age for the granite of 374 ± 2 Ma, in good agreement with a 10 point Rb–Sr isochron age of 360 ± 5 Ma reported by Irving and Strong (1985). Kerr et al. (1993) suggested that the granite was formed as an integral part of Appalachian-cycle magmatism and interpreted a prior four point Rb–Sr whole-rock isochron age of 315 ± 10 Ma (Bell and Blenkinsop 1975) to be disturbed and unreliable.

The St. Lawrence fluorspar deposits comprise fluorite veins that are found mostly within the granite or its related porphyry dikes (Fig. 2*b*). The veins are open fracture fillings that Collins and Strong (1988) believed were formed from regional stress and contraction during cooling of the granite. The fluorite is mostly massive and coarsely crystalline, forming veins from wall to wall. Close to the vein walls, the fluorite tends to be finely banded with rhythmic variations in colour. Two groups of veins have been recognized: (1) high-grade veins with strikes from 70° to 110° that average ≤1 m in width with acid-grade ore averaging ~95% CaF_2; and (2) low-grade veins with strikes from 140° to 160° that average ~7 m in width with grades up to ~70% CaF_2. The principal gangue minerals are quartz, cal-

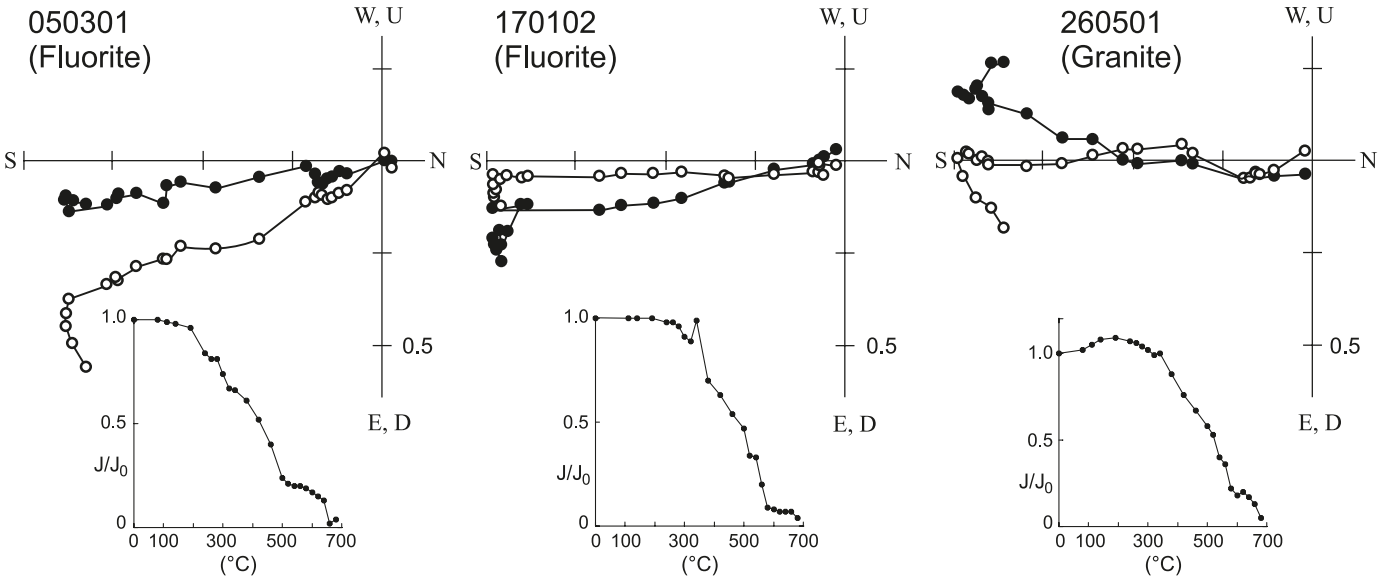
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Fig. 2. (*a*) Geological map of the St. Lawrence area with sampling sites (▲) of Irving and Strong (1984) (modified from Howse et al. 1983). (*b*) Geological setting of the fluorite veins at St. Lawrence with sampling sites (●) (modified from Howse et al. 1983; Strong et al. 1984).



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Fig. 3. Thermal step demagnetization of example specimens. Solid circles denote projection on the horizontal plane defined by the north (N), east (E), south (S), and west (W); and open circles denote projection on the vertical plane defined by the north (N), down (D), south (S), and up (U) axes. Axial intensity values (J) are expressed as a ratio of the natural remanent magnetization (NRM) intensity (J_0).



cite, sulphides, and barite with minor amounts of secondary iron, copper, and manganese minerals (Howse et al. 1983). Galena and sphalerite are the most abundant sulphides with pyrite and chalcopyrite occurring rarely.

Van Alstine (1948) concluded that the fluorite came from the same magma chamber as the granite and the ore fluids were released by late-stage differentiation, forming fluorite veins along faults and shears as the granite cooled and contracted. Similarly, Williamson (1956) concluded that the primary control on fluorite was the fracture pattern developed due to the stress system produced by cooling of the granite. From fluid inclusion, rare-earth element, and isotopic data, Strong et al. (1984) and Collins and Strong (1988) presented genetic models in which fluorite deposition occurred because of either increasing pH caused by boiling of magmatic fluid or mixing of cooled magmatic fluids with highly saline connate waters. Based on laser ablation – inductively coupled plasma – mass spectrometry (LA–ICP–MS) analyses, Gagnon et al. (2003) found that the fluorite exhibits compositional trends that represent either the progressive mixing of magmatic and formational waters or the interaction of hydrothermal fluid with wall rocks.

Methods and results

Sampling and initial measurements

Six to eight 2.54 cm diameter cores were drilled at 28 sites (Fig. 2*b*) and oriented in situ with a solar or magnetic compass, and three oriented block samples were collected at one additional site. The 29 sites included 19 sites in fluorite veins, 7 sites in granite, 1 site in a diabase dike, 1 site in the contact zone next to the dike, and 1 site in the contact zone of the granite's host rock. Six to 16 core specimens, each ~2.20 cm long, were prepared for each site in the paleomagnetic laboratory at the University of Windsor, Windsor, Ontario, providing 359 specimens. The specimens were stored for one month in a magnetically shielded room with an ambient field of $\sim 100\gamma$ to allow their unstable and undesirable viscous remanent magnetization (VRM) components to substantially decay. All subsequent magnetic measurements were done without removing the specimens from the shielded room. Natural remanent magnetizations (NRM) were measured on a 2G Enterprises 755R DC-SQUID magnetometer with a specimen sensitivity limit of $\sim 2 \times 10^{-6}$ A/m. The median NRM intensity of the fluorite specimens was 5.13×10^{-4} A/m (quartiles: $Q_1 = 1.35 \times 10^{-4}$, $Q_3 = 1.72 \times 10^{-3}$) and of the granite specimens was 3.19×10^{-3} A/m (quartiles: $Q_1 = 1.83 \times 10^{-3}$, $Q_3 = 3.98 \times 10^{-3}$).

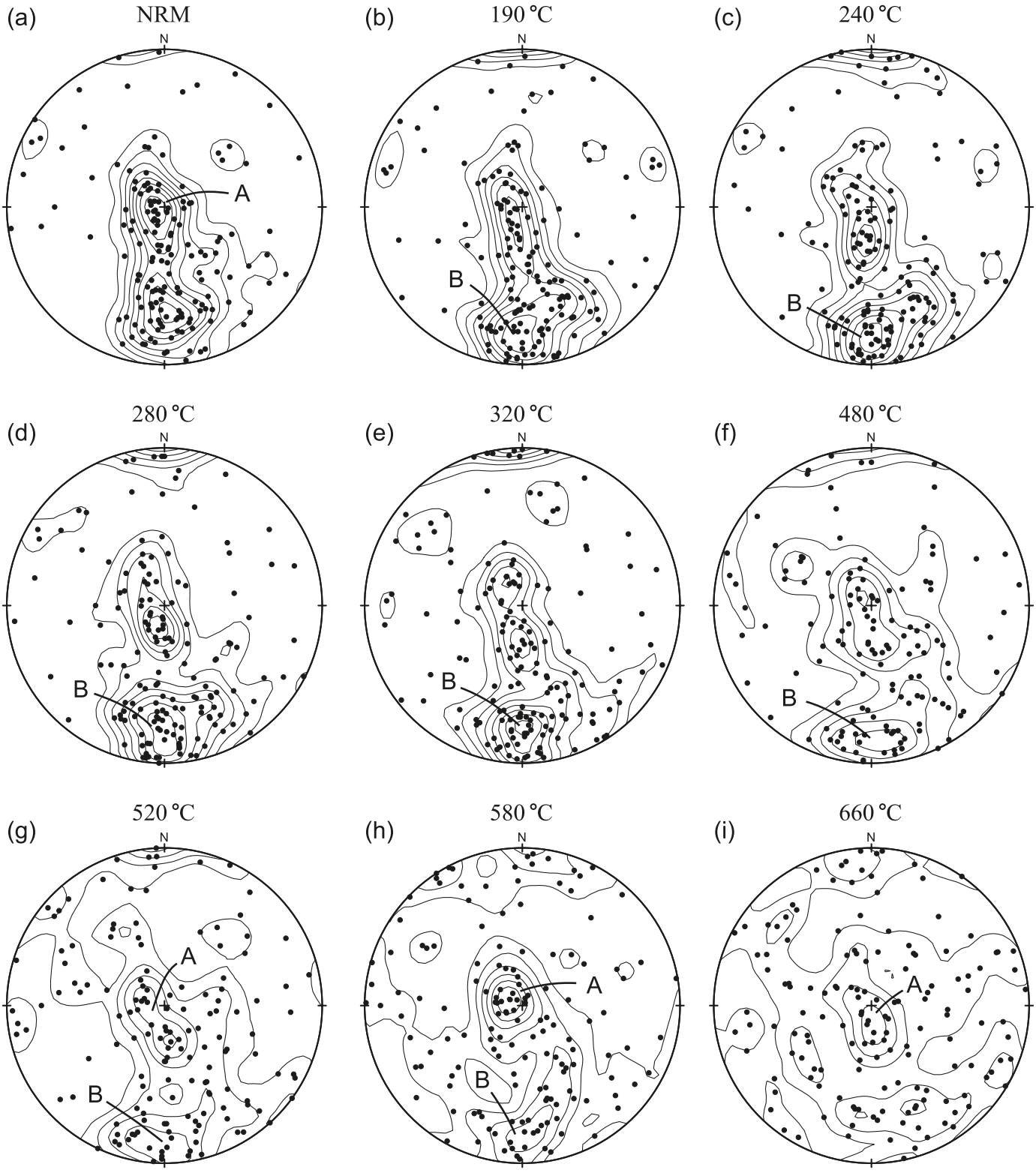
Demagnetization

One typical specimen from each site was thermally (Th) demagnetized in 24 steps to 680 °C using a Magnetic Measurements MMTD-80 oven with the Th steps biased into the diagnostic unblocking temperature ranges of the common magnetic minerals, i.e., goethite (80–140 °C), pyrrhotite (260–340 °C), magnetite (500–580 °C), and hematite (600–675 °C) (Dunlop and Özdemir 1997; Fig. 3). A second typical specimen from each site was alternating field (AF) demagnetized in 13 steps to 120 mT using a Sapphire Instruments SI-4 demagnetizer. Based on the initial results, the remaining specimens were either Th demagnetized in 19 steps up to 680 °C (Fig. 3) or AF demagnetized in 11 steps up to 120 mT. All remanence directions from the fluorite and granite specimens were plotted separately on Schmidt equal-area nets for each demagnetizing step using a computer program based on the Kamb method (1959), which gives directional density contours in 2σ increments (Fig. 4).

The fluorite specimens yielded two major clusters of NRM directions (Fig. 4*a*): (1) the A remanence component was directed almost vertically downward, apparently recording the present Earth's magnetic field; (2) the B component was directed shallowly downward to the south. During demagnetization, the A component cluster migrated towards

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Fig. 4. Vector directional density contour plots on equal-area projections of the lower hemisphere (Kamb 1959) for thermal step demagnetization data from the fluorite veins. Density contour intervals in 2σ increments. NRM, natural remanent magnetization.



the southerly shallow B component direction in steps up to ~280 °C (e.g., Figs. 4*b*–4*d*). Thereafter, the B component cluster was present and relatively constant in direction until ~580 °C (e.g., Figs. 4*d*–4*h*). Above ~580 °C, the B component directions started scattering rapidly (e.g., Fig. 4*i*). These results showed that the major remanence carriers are pyrrhotite and (or) magnetite. Similar results were observed in granite specimens. Therefore, the ChRM end-point directions for both vein and granite specimens belong to the B component cluster. Since the A component remains above

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Table 1. Site mean characteristic remanent magnetization (ChRM) directions.

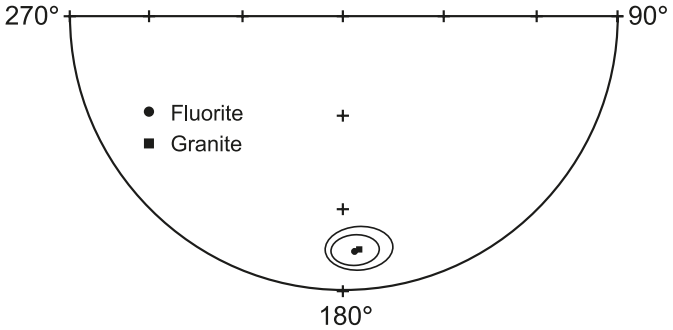
Site	Lithology	Strike (°)	Dip (°)	N^E	N^R	D (°)	I (°)	α_{95} (°)	k
1[a]	Fluorite vein (l)	335	87	—	—	—	—	—	—
2	Granite	—	—	3	5	182.2	12.3	9.9	36.2
3[a]	Fluorite vein (l)	125	75			—	—	—	—
4	Granite	—	—	7	0	171.4	20.1	8.0	57.8
5	Fluorite vein (h)	100	85	6	4	180.9	12.6	8.4	35.2
6	Fluorite vein (h)	70	85	10	5	175.8	13.4	7.9	40.8
7	Granite	—	—	9	0	166.8	19.2	7.7	45.4
8	Fluorite vein (h)	65	88	5	5	173.7	8.0	7.3	40.8
9	Granite	—	—	2	6	170.5	26.9	14.3	18.4
10	Fluorite vein (h)	107	90	1	3	197.8	16.8	9.3	180.5
11	Fluorite vein (h)	107	86	4	4	173.3	22.2	5.7	104.9
12	Granite	—	—	5	1	184.5	23.7	21.6	10.8
13	Fluorite vein (h)	105	88	3	4	161.7	7.8	17.4	15.4
14	Fluorite vein (h)	105	88	4	5	177.1	29.6	5.3	102.3
15	Fluorite vein (h)	85	81	4	3	169.1	0.9	9.0	52.2
16	Fluorite vein (h)	95	90	1	7	177.3	30.0	5.8	117.6
17	Fluorite vein (h)	280	90	8	1	180.0	15.7	7.5	48.5
18	Fluorite vein (l)	300	75	5	4	177.3	18.8	13.8	16.0
19[a]	Fluorite vein (h)	265	80	—	—	—	—	—	—
20	Granite contact	—	—	9	1	171.2	0.1	6.7	53.1
21	Fluorite vein (h)	110	90	1	6	169.4	21.8	10.2	46.9
22	Fluorite vein (h)	303	76	4	5	183.9	5.1	8.2	43.6
23[a]	Fluorite vein (h)	277	83	—	—	—	—	—	—
24	Diabase dike	107	72	1	11	152.3	49.8	5.4	74.3
25	Dike contact	—	—	15	0	121.5	–7.7	4.7	67.6
26	Granite	—	—	10	0	187.4	11.6	10.3	22.8
27	Fluorite vein (l)	293	83	10	0	172.8	4.0	14.9	11.4
28	Fluorite vein (l)	293	83	6	0	186.2	16.9	8.0	59.8
29	Granite	—	—	5	1	176.1	25.3	5.7	142.6
RDH	Granite	—	—	—	—	172	2	9	27
SLH	Granite	—	—	—	—	174	–4	19	11
Mean	Fluorite	—	—	15	—	177.0	15.0	5.7	46.0
	Granite[b]	—	—	10	—	175.6	13.8	7.8	39.3

Note: High-grade (h) and low-grade veins (l) are determined by their strike and width. Mean declination (D), inclination (I) and radius of cone of 95% confidence (α_{95}) in degrees (°) and precision parameter (k) of Fisher (1953). N^E, specimen end points; N^R, remagnetization circles. RDH, Redhead (H component), SLH, St. Lawrence village (H component) from Irving and Strong (1985).

[a]Omitted sites from average directions due to incoherent specimen directions.

[b]Includes the baked contact (site 20), SLH, and RDH sites.

Fig. 5. Unit mean characteristic remanent magnetization (ChRM) directions on the lower hemisphere of an equal-area stereonet.

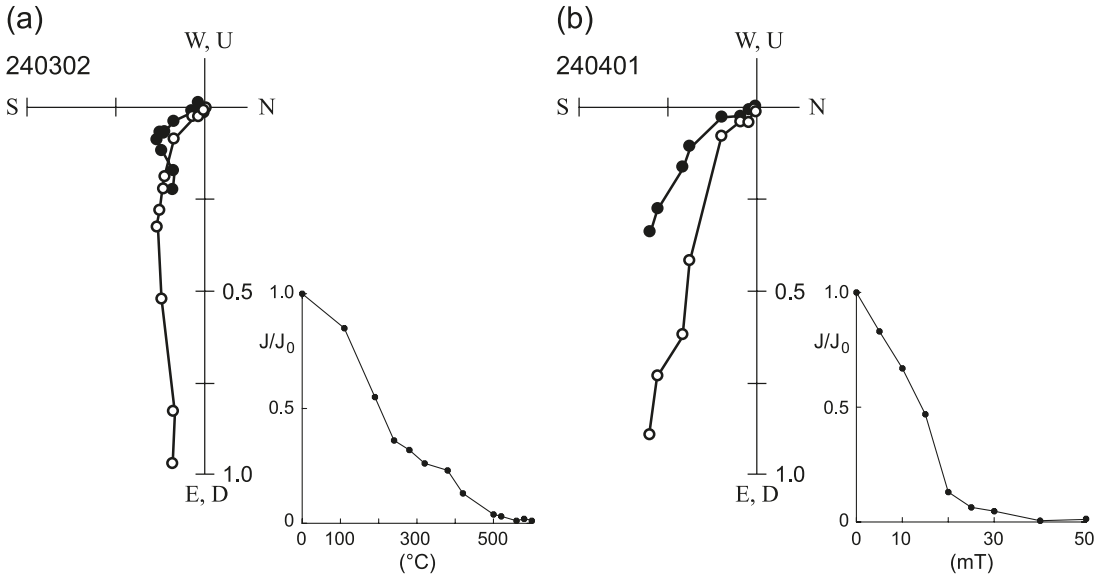


580 °C in some specimens, its ChRM carrier is hematite that was generated mostly by modern weathering.

The specimen ChRM directions were determined using principal component (Kirschvink 1980) and remagnetization circle analysis (Bailey and Halls 1984), thereby isolating the A and B component directions from the total population of remanence directions. Site and unit mean directions were calculated following Fisher (1953) (Table 1). The site mean ChRM directions were dispersed but mostly in the B component direction (Fig. 5). The mean direction for 15 sites in fluorite veins is declination (D) = 177.0° and inclination (I) = 15.0° (number of sites (N) = 15, radius of cone of 95% confidence (α_{95}) = 5.7°, precision parameter of Fisher (1953) (k) = 46.0), giving a paleopole at 35.4°N, 128.2°E (with semi-axes for the oval of 95% confidence of dp = 3.0° and dm = 5.8°). Including the two sites from Irving and Strong (1985) (Redhead (H component) and St. Lawrence village (H component) sites), the mean direction for nine granite sites is D = 176.1° and I = 15.4° (N = 9, α_{95} = 8.1°, k = 41.2), giving a paleopole at 35.2°N, 129.2°E (dp = 4.3°, dm = 8.3°). The Redhead H component direction was derived from only 11 specimens from nine samples spaced

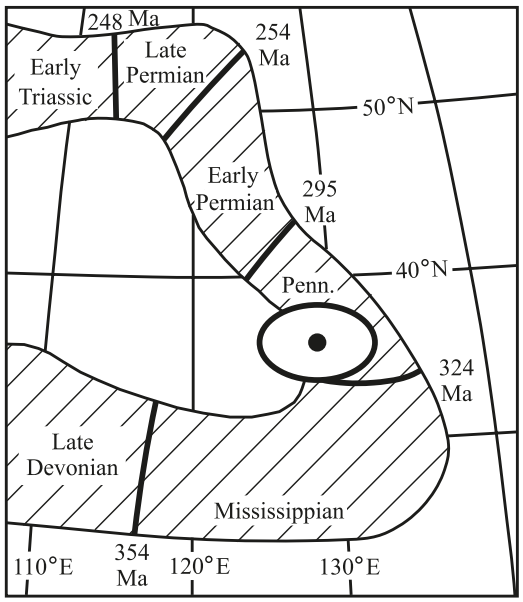
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Fig. 6. Step demagnetization of example specimens of the dike: (*a*) thermal step demagnetization; (*b*) alternating field step demagnetization. Plotting conventions as in Fig. 3.



at intervals up to 1 km apart around the shoreline, and the St. Lawrence H component from seven specimens from seven similarly spaced samples along Highway 220 across the batholith (Irving and Strong 1985).

Fig. 7. Pole position for the combined population of fluorite vein and granite site mean directions with its oval of 95% confidence (Table 2) on the apparent polar wander path (APWP) for North America of Van der Voo (1993). Penn., Pennsylvanian.



Contact test

A paleomagnetic baked contact test is useful for determining whether or not the studied lithology has undergone pervasive secondary remanence overprinting. If an igneous intrusion and its baked contact zone give a ChRM direction that differs from that of remote host rock, then the baked contact test is positive and shows that the host rock's ChRM was acquired before the dike was intruded (Everitt and Clegg 1962). Conversely, if the dike, baked contact zone, and host rock give the same ChRM direction, then the test is negative and indicates that the host rock was remagnetized at the time of intrusion or later. In this study, we attempted three contact tests.

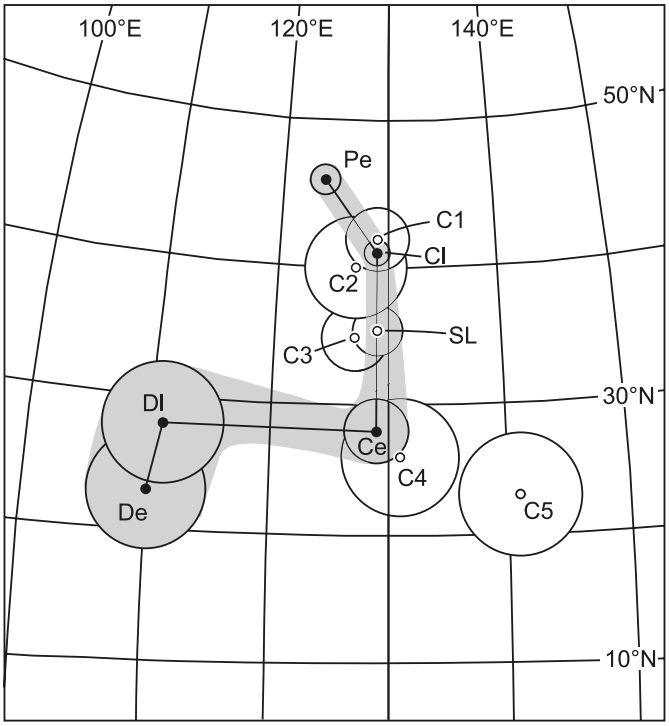
Site 3 is in a ~1.4 m wide fluorite vein (strike 125°, dip 75°S), and site 4 is in its granitic contact zone. Although the ChRM directions of site 3 are scattered, the site mean ChRM direction for the contact zone is not significantly different at 95% confidence from the mean direction for the B component in the 15 fluorite vein sites (Table 1). This result indicates that the ChRM in the fluorite veins and the granite are coeval in origin.

Site 24 is in a ~2 m wide diabase dike (strike 107°, dip 72°S), site 25 is in the dike's baked contact zone, and site 26 is in remote host rock about 35 m from the dike. The mean ChRM direction of the dike is D = 152.3° and I = 49.8° (N = 12, α_{95} = 5.4°, k = 74); and it is significantly different from both the mean ChRM direction of its contact zone of D = 121.5° and I = –7.7° (N = 15, α_{95} = 4.7°, k = 68) and its remote host rock, which gives the B component direction. The dike's virtual pole position is at 8.6°N, 148.4°E (dp = 4.8°, dm = 7.2°). Thermal step demagnetization of the dike's specimens indicates that the main remanence carrier is magnetite (Fig. 6*a*). Also, the rapid NRM intensity decay on AF step demagnetization results for the dike specimens indicate that the magnetite remanence is carried by both single domain (SD) or pseudosingle domain (PSD) grains and multi-domain (MD) grains (Fig. 6*b*). Such MD magnetite is likely to carry a VRM that easily changes direction to the ambient Earth's magnetic field to give the aberrant direction.

Site 20 sampled the Cambrian Inlet Group (Fig. 2*b*) in the St. Lawrence granite's contact zone. The site 20 mean direction is not significantly different from the unit mean direction of the granite (Table 1), suggesting that the granite's ChRM has been stable since it was intruded. Since there is no significant difference between the mean directions of

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Fig. 8. Carboniferous paleomagnetic poles from the northern Appalachians (C1, C2, C3, C4, and C5) and St. Lawrence granite and fluorite veins (SL) with their cones of circular standard deviation (1σ confidence) (Table 2). The APWP for North America is represented by a solid line with its 1σ confidence bounds (shaded area) from Van der Voo (1993). Ce, Early Carboniferous; Cl, Late Carboniferous; De, Early Devonian; Dl, Late Devonian; Pe, Early Permian.



site 20 and the nine granite sites, the ChRM direction of site 20 can be combined to recalculate the unit mean direction of the granite (D = 175.6°, I = 13.8°, N = 10, α_{95} = 7.8°, k = 39.3).

Discussion

There is no statistically significant difference at 95% confidence between the unit mean ChRM directions of the fluorite veins and granite batholith (correlation test; f = 0.3837 << F = 3.232 (2σ), McFadden and Jones 1981). Thus, the site mean directions for the fluorite veins (15 sites), granite (nine sites), and the granite contact (site 20) can be combined, giving a mean ChRM direction for the B component of D = 176.5° and I = 14.6° (N = 25, α_{95} = 4.4°, k = 44.7) (Fig. 5) with a pole position at 35.6°N, 128.8°E (dp = 2.3°, dm = 4.5°). This paleopole falls on the Early Pennsylvanian (Gradstein and Ogg 2004) portion of the apparent polar wander path (APWP) for North America of Van der Voo (1993) (Fig. 7). However, the implied paleomagnetic age of 316 ± 8 Ma is ~58 Ma younger than the granite's Late Devonian U–Pb zircon age of 374 ± 2 Ma. There are three possibilities to consider for the origin of this implied age: (1) a Kiaman overprint, (2) remagnetization by later hydrothermal fluids forming the fluorite veins, and (3) tectonics.

The Kiaman superchron represents a 50–60 Ma interval in the Pennsylvanian and Permian during which the Earth's magnetic field maintained a reversed polarity (Buchan and Chandler 1999). A Kiaman overprint has been observed in many North American and European Paleozoic sedimentary rocks (e.g., Kent and Opdyke 1978; Van der Voo et al. 1979; McCabe and Elmore 1989). Irving and Strong (1985) interpreted their H remanence component in the St. Lawrence granite to be a Kiaman overprint from low temperature alteration that deposited hematite diffusely throughout the granite and their N component to be the primary magnetization acquired during cooling of the pluton. However, their N component direction is from only 22 specimens and gives an inconsistent Early Silurian paleopole at 11.1°N, 117.6°E (dp = 9.1°, dm = 13.5°) that predates the Late Devonian St. Lawrence granite. Therefore, it is highly unlikely that their N component is a valid primary component. Further, the main remanence carriers of the fluorite veins and granite specimens are pyrrhotite and (or) magnetite with only minor hematite. If low temperature alteration had caused hematite deposition in the rocks as Irving and Strong (1985) interpreted for their H component, the hematite remanence should be dominant and its direction should be the similar to the B component, not the observed low-intensity A component direction of this study. Excluding the results from sites SLH and RDH in the St. Lawrence granite of Irving and Strong (1985), we offer no explanation for the origin of their N and H components, which were obtained from several rock units in addition to the granite. The slight difference between the present Earth's magnetic field direction in the study area (average 1900–2005 is D = 334°, I = 71°; IGRF-10, International Geomagnetic Reference Field) and the hematite A component is likely the result of the combination of a large secondary modern weathering component in hematite and a small primary hematite component.

Remagnetization of the granite in a later thermal event by the hydrothermal fluid that formed the fluorite veins is highly unlikely. Strong et al. (1984) and Collins and Strong (1988) presented a genetic model based on geochemical observations, including fluid inclusion, rare earth element, and isotopic data in which the fluorite veins were formed from late-stage hydrothermal fluids that were derived from the granite magma. In this study, the fluorite vein specimens were collected from both the narrow, high-grade E–W and thick, low-grade NNW–SSE veins; and the granite specimens were collected from both close to and remote from the veins. There is no significant difference between ChRM directions for the granite specimens that are close to, or remote from, the veins, indicating no impact by later hydrothermal fluid. Further, the peak homogenization temperature (T_p) found in the fluorite fluid inclusions is ~485 °C (Strong et al. 1984). Since fluorite responds like calcite to a metamorphic event (Turgarinov and Vernadsky 1970; Bodnar and Bethke 1984; Goldstein and Reynolds 1994), a lower amphibolite-facies regional metamorphic event would be required to have re-equilibrated the fluorite to give a T_p of 485 °C (Barker and Goldstein 1990), whereas the granite and surrounding country rocks provide no evidence of a metamorphic event (Strong et al. 1978). Also, it is unlikely that the small veins could carry sufficient heat to remagnetize the entire granite batholith. Thus, it is highly unlikely that the granite was remagnetized by later hydrothermal fluid.

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Table 2. Carboniferous paleomagnetic poles from the northern Appalachians and cratonic North America.

Key[a]	Rock unit	Lithology	Latitude °N	Longitude °E	A_{95}°	References
Late Carboniferous[b]						
C1	Combined pole of Maritimes[c]	—	42	129	3.8	Johnson and Van der Voo (1989)
C2	Combined pole of craton	—	40	127	6.1	Johnson and Van der Voo (1989)
mid-Carboniferous[b]						
C3	Combined pole of Maritimes[c]	—	35	121	4.2	Johnson and Van der Voo (1989)
Early Carboniferous[b]						
DL	Deer Lake Group, NL	Red to grey sandstones and siltstones	21	128		Irving and Strong (1984)
T	Terrenceville, NL	Red sandstone	27	124		Kent (1982)
SF	Sprout Falls, NL	Red sandstone	29	140		Murthy (1985)
JV	Jeffreys Village, NL	Red sandstone and siltstone	27	131		Murthy (1985)
FB	Fisset Brook, NS	Red beds and volcanics	20	139		Johnson and Van der Voo (1989)
NB	New Brunswick, NB	Red beds and volcanics	21	135		Seguin et al. (1985)
CH	Cheverie Formation, NS	Red sandstone	24	152		Spariosu et al. (1984)
WR	White Rock Formation, NS	Sandstone and volcanics	22	148		Spariosu et al. (1984)
ST	Stellarton Graben, NS	Sandstone and conglomerate	29	131		Morris (2002)
C4	Combined pole of NL	Key poles: DL, T, SF, and JV	26	131	7.9	
C5	Combined pole of NB and NS	Key poles: FB, NB, CH, WR, and ST	23	141	8.3	
Devonian						
SL	St. Lawrence	Granite and fluorite veins	36	129	3.4	This study

[a]The symbol is used to identify collection locations and poles in Figs. 1 and 8.

[b]Early Carboniferous (Tournaisian and Viséan), mid-Carboniferous (Namurian), Late Carboniferous (Westphalian and Stephanian).

[c]Martimes includes poles from the northern Appalachian orogen in New Brunswick (NB), Newfoundland (NL), and Nova Scotia (NS), Canada.

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Kent and Opdyke (1978) reported that upper Paleozoic sedimentary basins within the Appalachian orogenic belt gave paleomagnetic poles that were divergent from those from ancestral North America. They suggested the discrepancy was evidence for large-scale (~1500 km) lateral translation of the north-central Appalachians (Acadia) during the mid-Carboniferous. However, Irving and Strong (1984) reported that the Early Carboniferous sedimentary rocks of the Deer Lake Group in Newfoundland (Fig. 1) gave a paleopole that was not statistically different from other paleopoles within the orogenic belt, and they concluded that such a large-scale translation did not occur during or after Viséan time (345–326 Ma). Subsequent studies (Johnson and Van der Voo 1989; Morris 2002) supported Kent and Opdyke's (1978) original observation, concluding that paleopoles from locations within the orogenic belt are significantly different from those sampled on the craton.

The fluorite veins sampled in this study come from both E–W-trending, high-grade veins and NNW–SSE-trending, low-grade veins. The average dip of the veins is almost vertical (dip azimuth = 44.4°, dip angle = 88.9° (N = 19, α_{95} = 3.2°, k = 114)). Further, there is no significant difference between the mean ChRM directions of the high-grade veins (D = 176.1°, I = 16.0°, N = 10, α_{95} = 6.6°, k = 55) and low-grade veins (D = 178.7°, I = 13.3°, N = 3, α_{95} = 16.0°, k = 60), which makes post-emplacement tilting of the St. Lawrence granite improbable. The U–Pb zircon age of 374 ± 2 Ma for the St. Lawrence granite (Kerr et al. 1993) corresponds to a pole position on the North American APWP of 30°N, 110°E (radius of cone of 95% confidence, A_{95} = 13°) (Van der Voo 1993). To match this pole position, the declination for the ChRM of the St. Lawrence fluorite veins and granite combined (25 sites) must be rotated clockwise by 17° ± 11°, indicating at least that this part of the Avalon Terrane of Newfoundland was rotated counterclockwise by 17° ± 11° after the granite and fluorite veins were emplaced. The relatively large error for the rotation estimate is caused by the large A_{95} value of the reference pole.

The timing of the rotation of the terrane can be determined by comparing northern Appalachian to cratonic North American paleopoles. Figure 8 is a plot of Carboniferous paleomagnetic poles determined from the northern Appalachians. Note that the Late (C1, C2) and middle (C3) Carboniferous paleopoles (Table 2) are not significantly different from the North American reference pole path of Van der Voo (1993). However, the Early Carboniferous paleopole from New Brunswick and Nova Scotia (C5) is displaced from its reference pole, whereas the paleopole from Newfoundland (C4) is not significantly different from the reference pole (Fig. 8). This result indicates that, while the mainland provinces may have been rotated or displaced, Newfoundland has not undergone significant displacement with respect to stable North America since the beginning of the Carboniferous. Therefore, the counterclockwise rotation of the Avalon Peninsula occurred during Late Devonian and was probably related to the Acadian orogeny. The Acadian orogeny is generally thought to be Devonian with the upper age limit for Acadian deformation provided by the Viséan Codroy Group cover rocks (Waldron et al. 1998).

The paleolatitude found for the Late Devonian St. Lawrence paleopole of 7.4° ± 4.4° is the same at 95% confidence as the 12° ± 10° value for the Late Devonian cratonic paleopole of Van der Voo (1993). The paleolatitudinal difference between the St. Lawrence paleopole and the Early Carboniferous poles (Fig. 8) is attributed in part to apparent polar wander and in part to inclination shallowing in the sedimentary rocks and (or) the presence of modern hematite rather than to latitudinal displacement of the terrane. The Early Carboniferous data are collected from sandstones, red beds, or both red beds and volcanics (Table 2). Shallowing of the inclination causes an underestimate of the paleolatitude. Tan and Kodama (1998) reported that clay-containing sedimentary rocks typically experience from 10° to 20° of inclination shallowing due to burial compaction. Based on the remanence intensity for redeposited sediments as a function of the salinity and pH of the sediment–water mixture, Katari and Tauxe (2000) reported that magnetite is always included as a combined clay–magnetite assemblage that is the smallest unit reoriented by a magnetic field. Also, anomalously shallow inclinations are found worldwide in red beds (e.g., Li et al. 1988; van der Pluijm et al. 1993; Garcés et al. 1996). Recently, Tan et al. (2007) reported rock magnetic results that indicate inclination shallowing in red beds from the northern Appalachian Newark basin has caused a systematic error in the APWP. In contrast, the St. Lawrence samples come from fluorite veins and granite that are not affected by inclination shallowing, that are shown by vein attitudes to be untilted, and in which the main remanence carriers are pyrrhotite and (or) magnetite rather than hematite.

Murthy (1983) reported paleomagnetic results for the Early Devonian Deadman's Bay dikes in the Gander Zone in Newfoundland (Fig. 1) and suggested that any relative motion between Acadia and cratonic North America predated the Early Carboniferous (Mississippian). Gales et al. (1989) reported paleomagnetic results from Silurian rocks in the Dunnage Zone and concluded that the Central Mobile Belt of Newfoundland was adjacent to the North American craton. However, they also indicated the possibility of a small (<10°) counterclockwise rotation of central Newfoundland. These results are consistent with our St. Lawrence paleopole in suggesting that the Gander, Dunnage, and Avalon zones underwent a minor pre-Carboniferous counterclockwise rotation of ~17° during the Acadian orogeny.

Conclusions

There is no statistically significant difference at 95% confidence between the site mean ChRM directions of the St. Lawrence fluorite veins and granite. Since no post-granite metamorphic event is known in the area, these concordant directions indicate that the granite and fluorite veins are coeval in origin and retain a primary remanence. This syngenetic result agrees with a genetic model in which the fluorite veins were formed from late-stage hydrothermal fluids that were derived from the granite magma (Strong et al. 1984; Collins and Strong 1988). However, the pole position of St. Lawrence fluorite and granite falls on the Early Pennsylvanian portion of the APWP for North America, which is much younger than the granite's Devonian age. The declination difference between the measured and expected paleopoles is explained by a postemplacement counterclockwise

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rotation of the St. Lawrence granite region about a vertical axis by $\sim 17^\circ \pm 11^\circ$. The timing of this rotation at St. Lawrence, from the analysis of Carboniferous paleopoles in the northern Appalachians, is Late Devonian during the Acadian orogeny.

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